

torch.nn.functional.linear#

<https://pytorch.org/docs/stable/generated/torch.nn.functional.linear.html>

Description:

Applies a linear transformation to the incoming data: $y = xA^T + b$. This operation supports 2-D weight with sparse layout. Sparse support is a beta feature and some layout(s)/dtype/device combinations may not be supported, or may not have autograd support. If you notice missing functionality please open a feature request. This operator supports TensorFloat32.

Function Signature

```
torch.nn.functional.linear(input, weight, bias=None) → Tensor#
```

Notes & Warnings

WARNING

Warning Sparse support is a beta feature and some layout(s)/dtype/device combinations may not be supported, or may not have autograd support. If you notice missing functionality please open a feature request.

Detailed Documentation

Documentation

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This operation supports 2-D weight with sparse layout

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This operator supports TensorFloat32.

Input: $(*, in_features)(*, in_features)(*, in_features)$ where $*$ means any number of additional dimensions, including none

Weight: $(out_features, in_features)(out_features, in_features)(out_features, in_features)$ or $(in_features)(in_features)(in_features)$

Bias: $(out_features)(out_features)(out_features)$ or $()()$

Output: $(*, out_features)(*, out_features)(*, out_features)$ or $(*)(*)(*)$, based on the shape of the weight